# Section 3.10 Linearization and Differentials

Ex ① Find the equation of the tangent line for $f(x) = \sqrt{x}$ at $x = 4$.

We need to find $f(4) = \sqrt{4} = 2$ so that we have the point $(4,2)$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

This means we have $m = \frac{1}{4}$ at $(4,2)$.

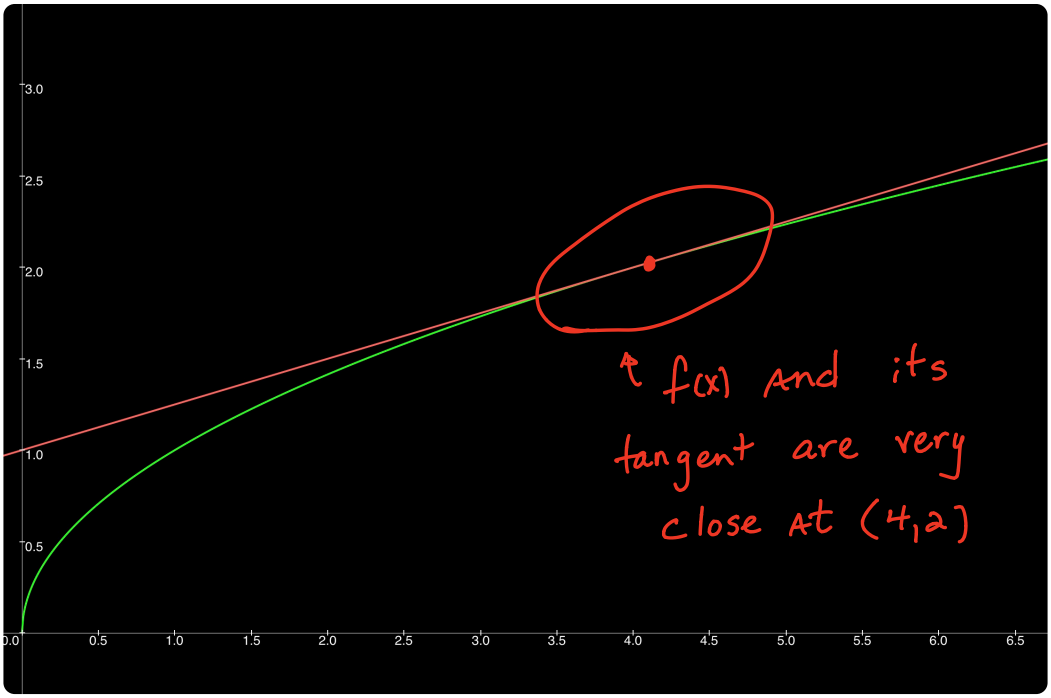
Use point slope formula.

$$y - 2 = \frac{1}{4}(x - 4)$$

$$y - 2 = \frac{1}{4}x - 1$$

$$y = \frac{1}{4}x + 1$$

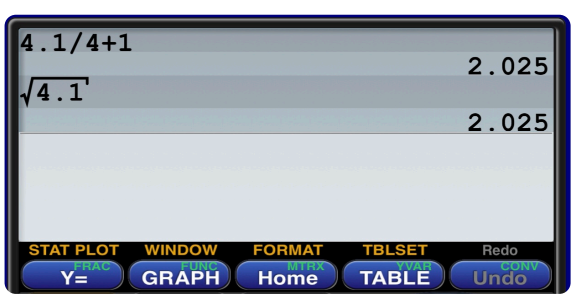
Now graph the original function and the tangent line to see if they look correct.




Next, use $x = 4.1$ in the tangent line.

$$y = \frac{1}{4}(4.1) + 1$$

$$y = 2.025$$




estimate

Actual Answer

This means we can use the tangent line to estimate $\sqrt{x}$, as long as $x$ is close to 4!

This process is call linearization.

The idea is to be able to calculate complicated functions with easy to use estimates — even without a calculator.

Ex ② Use linearization to estimate $\sin(3.2)$.

Start with $y = \sin x$ and pick a number close to 3.2 that you can find sine of without a calculator.

use $x = \pi$

then $y = \sin(\pi) = 0$

So the point is $(\pi, 0)$ and next we take the derivative.

$$y' = \cos x$$

$$y'(\pi) = \cos(\pi) = -1$$

Use the point slope formula with $m = -1$ at $(\pi, 0)$.

$$y - 0 = -1(x - \pi)$$

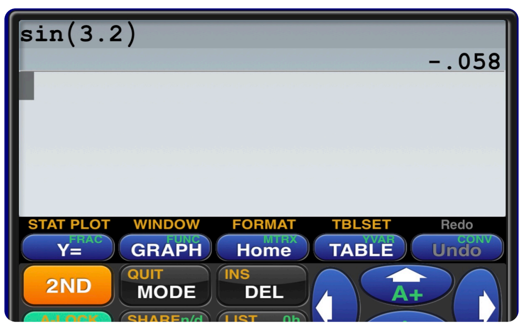
$$y = -x + \pi$$

Finally, substitute $x = 3.2$

$$y = -3.2 + \pi$$

$$y = -0.058$$

Compare that with the actual answer.

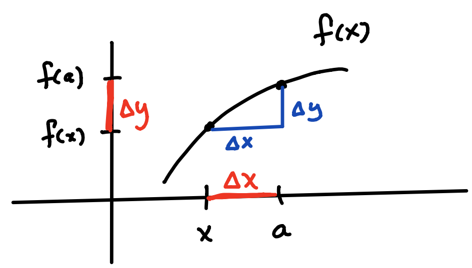



So, the simple line $y = -x + \pi$ can be used to estimate $y = \sin x$ as long at $x$ is close to $\pi$.

# Differentials: dy And dx

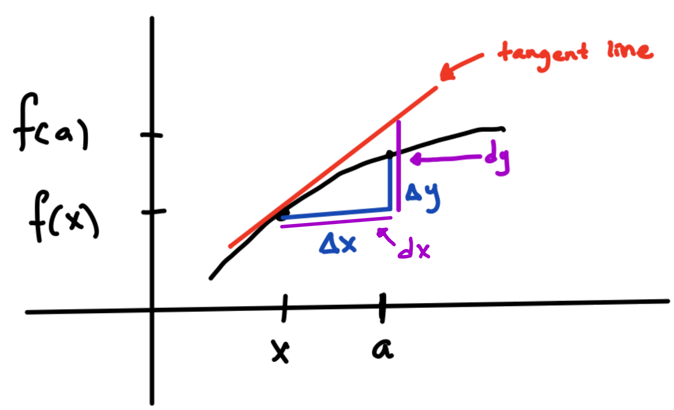
We often use the notation $y'$ in place of $\frac{dy}{dx}$, but what are dy and dx?

Let's go back to the definition of the derivative.




$$\text{Slope} = \frac{f(x) - f(a)}{x - a} \quad \text{or} \quad \frac{\Delta y}{\Delta x}$$

And $$\frac{dy}{dx} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$



This means that $dx = \Delta x$ and $dy \approx \Delta y$.

$\Delta x$ and $\Delta y$ are the change of $x$ and $y$.

$$\Delta x = x_2 - x_1 \quad \text{and} \quad \Delta y = y_2 - y_1$$

$dx$ and $dy$ are the differentials, which are found by finding the derivative $\frac{dy}{dx}$.

<u>Ex</u> 3 $y = x^2$ is the area of a square.

Calculate $\Delta x$, $\Delta y$, $dx$ and $dy$ if the side of the square changes from 8 to 8.1

$$\Delta x = 8.1 - 8 = 0.1$$

<u>use</u> $dx = \Delta x$

$$dx = 0.1 \text{ also.}$$

$$x = 8 \rightarrow y = 8^2 = 64$$

$$x = 8.1 \rightarrow y = 8.1^2 = 65.61$$

$$\Delta y = 65.61 - 64$$

$$\Delta y = 1.61$$

This means that if the sides grow by 0.1 units, then area increases by 1.61 units$^2$.

$dy$ will be close to 1.61, so we can use $dy$ to estimate how much area changes.

$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$dy = 2x \cdot dx \quad \text{substitute } x=8,\ dx=0.1$$

$$dy = 2(8)(0.1) = 1.6$$

Notice that $\Delta y = 1.61$ and $dy = 1.6$ are very close.

Ex ④ A spherical balloon is supposed to be inflated to a radius of 12 inches, but it is possible that the radius could be off by $\pm 0.2$ inches.

Use differentials to estimate how much the Volume could be off by.

$$V = \frac{4}{3}\pi \cdot r^3 \quad \text{And} \quad dr = \pm 0.2$$

$$\frac{dV}{dr} = 4\pi r^2$$ solve for dV

$$dV = 4\pi r^2 \cdot dr$$

$$dV = 4\pi (12)^2 (\pm 0.2)$$

$$dV = \pm 361.911 \text{ in}^3$$

Notice that the volume should be

$$V = \frac{4}{3}\pi (12)^3 = 7238.229 \text{ in}^3$$

So, An error in volume of $\pm 361.911$ in$^3$

is A % error of $\pm \frac{361.911}{7238.229} \times 100$

$$= \pm 5\%$$